

Rapid detection of *Macrophomina phaseolina* in common bean seeds using a visual loop-mediated isothermal amplification assay

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Abstract *Macrophomina phaseolina* (Tassi) Goid is a pathogen that causes significant economic losses by infesting *Phaseolus vulgaris* L. (common bean). Here, we developed a rapid, visual, and sensitive loop-mediated isothermal amplification (LAMP) assay for *M. phaseolina* detection in common bean seeds based on sequence-characterized amplified regions (GU018142.1). The ideal conditions for detection were obtained in 45 min at 65 °C. The LAMP assay was visually inspected for the addition of hydroxynaphthol blue dye. Positive reactions turned sky blue while the negative controls remained violet under ambient light. The detection limit was 1 pg of *M. phaseolina* genomic DNA per reaction, and no cross reactions were observed with DNA isolated from five other seedborne pathogens. The LAMP assay detected both common bean seeds naturally infected by *M. phaseolina*, and seeds artificially contaminated with the pathogen. The limit of detection was one infected seed per lot of 400 seeds. This detection method was possible without prior fungal DNA extraction, by soaking seeds 2–6 h. These results suggest that the LAMP assay provides a rapid qualitative diagnostic tool for detecting *M. phaseolina* in bean seeds.

Keywords Detection · Hydroxynaphthol blue · Molecular diagnosis · Seed pathology

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Introduction

Macrophomina phaseolina (Tassi) Goid is an important fungal plant pathogen that causes charcoal rot of the common bean (*Phaseolus vulgaris* L.). *M. phaseolina* infects, in total, more than 500 plant species, including key agricultural crops like cotton, soybeans, maize, sorghum, and sunflower (Dhingra and Sinclair 1978). The fungus can remain viable for more than 4 years in soil and crop residue as sclerotia, which makes it difficult to control (Short et al. 1980).

M. phaseolina is a common soil and seedborne pathogen, found both on and inside the infected seed (Abawi and Pastor-Corrales 1990). Transmission of the pathogen via infected seeds is known to be a means of spreading *M. phaseolina* into new areas. Another adverse effect of this seedborne pathogen is that infected seeds may fail to germinate, or they may produce infected seedlings that can transmit the pathogen and augment the inoculum potential in the soil. An important preventive management tool for reducing the spread of pathogens into new regions is accurate seed health testing.

The phytosanitary seed-health testing protocols currently used for *M. phaseolina* detection begin with a conventional blotter test, followed by either a grow-out test (ISTA 1976) or growth on a semi-selective medium (Mihail and Alcorn 1982). Both techniques are time consuming and require mycological skills, hampering their large scale usage, and even when performed correctly, they are not always sensitive enough to detect low levels of seed infection.

Molecular tools have greatly facilitated the unequivocal identification of fungal pathogens (Kaur et al. 2012; Niessen 2015). In this context, polymerase chain reaction (PCR)-based methods for the detection of *M. phaseolina* have been described based on ribosomal gene sequences (Babu et al. 2007; Biswas et al. 2014). The detection and quantification of *M. phaseolina* from soil and plants with real-time

quantitative PCR using TaqMan and SYBR Green assay techniques have been developed by Babu et al. (2011).

Recently, loop-mediated isothermal amplification (LAMP) of DNA has emerged as a promising alternative to PCR for the detection of many pathogenic fungi, such as *Sclerotinia sclerotiorum* (Duan et al. 2014a), *Botrytis cinerea* (Tomlinson et al. 2010; Duan et al. 2014b), *Phytophthora sojae* (Dai et al. 2012), and *Fusarium graminearum* (Niessen and Vogel 2010), *Fusarium oxysporum* f. sp. cubense (Zhang et al. 2013), *Fusarium oxysporum* f. sp. niveum (Peng et al. 2013), *Fusarium oxysporum* f. sp. *Ciceris* (Ghosh et al. 2015).

The LAMP method can amplify nucleic acids with high specificity, sensitivity, and rapidity under isothermal conditions (Notomi et al. 2000). The LAMP reaction uses a set of two specially designed inner primers (FIP and BIP) and two outer primers (F3 and B3), which are designed to amplify 6 or 8 distinct regions of the target DNA. Positive LAMP reactions can be monitored with the naked eye by adding DNA-intercalating dyes such as ethidium bromide, SYBR Green I, propidium iodide or Quant-iT PicoGreen; by adding metal-ion indicators such as hydroxynaphthol blue (HNB; Goto et al. 2009) or calcein (Tomita et al. 2008); or by measuring the increase in turbidity caused by a white precipitate from magnesium pyrophosphate (Mori et al. 2001; Tomita et al. 2008). The reaction can also be monitored in real-time, allowing for quantitative detection of the target (Mori et al. 2004; Mori and Notomi 2008; Goto et al. 2009; Tomlinson et al. 2010; Zhang et al. 2013).

In this current study, a LAMP assay was evaluated based on the use of sequence characterized amplified region (SCAR) with HNB for a rapid visual detection of *M. phaseolina* present in common bean seeds.

Materials And methods

Fungal isolates and DNA extraction

The pathogen *M. phaseolina*, isolate Mph 22S/2014 (Table 1) was isolated from infected common bean (*Phaseolus vulgaris* L.) seeds. Isolates of *Sclerotinia sclerotiorum*, *Fusarium oxysporum* f. sp. *phaseoli*, *Rhizoctonia solani* and *Trichoderma harzianum* (Table 1) obtained from the Federal University of Goiás Collection, were originally collected from Goiás State between 2014 and 2015. Fungal isolates were maintained at 28 ± 1 °C on potato dextrose agar (PDA; Difco Laboratories) with 0.1 g L^{-1} ampicillin (Sigma-Aldrich, St. Louis, MO).

For isolation of DNA, a 5-mm disc of 2-day-old culture was grown for 3 days at 28 ± 1 °C in a 150 ml Erlenmeyer flask containing 50 ml potato dextrose broth (pH 5.5) with 0.1 g L^{-1} ampicillin (Sigma-Aldrich, St. Louis, MO). The

mycelia were filtered through Whatman No.1 filter paper and DNA was isolated by the cetyltrimethylammonium bromide (CTAB) method as described by Doyle and Doyle (1987). Quality and concentration of DNA were tested using a NanoDrop ND-1000 Spectrophotometer (Thermo Fischer Scientific, Wilmington, DE, USA), followed by gel electrophoresis (6 V.cm^{-1}) on 1% agarose (Invitrogen, Carlsbad, CA) in 1X TBE buffer (Tris-borate-EDTA) containing $0.5 \text{ }\mu\text{g.mL}^{-1}$ ethidium bromide and viewed under UV transilluminator.

Seed inoculation with *Macrophomina phaseolina*

Four hundred seeds of the common bean (*Phaseolus vulgaris* L. cv. BRS Pérola), were artificially inoculated with an isolate of *M. phaseolina* (Mph 22S/2014, Table 1) using the water restriction technique (Machado et al. 2004). Briefly, seeds were surface disinfested by soaking for 2 min in a solution containing 1% NaHClO, followed by complete drying on a sterile filter paper under a laminar flow hood. The isolate Mph 22S/2014 was cultured for 5 days on PDA medium (Difco Laboratories; supplemented by mannitol with the osmotic potential adjusted to -1.4 MPa) at 28 ± 1 °C, and then seeds were placed on the fungal colonies. The plates were closed, wrapped with foil and incubated for 3 days at 28 ± 1 °C and the seeds were then air dried overnight under a laminar flow hood.

Artificial infection levels, ranging from 0% to 40% seed infection incidence, were created by mixing *M. phaseolina*-inoculated seeds with non-infected seeds (0/400; 1/400; 2/400; 4/400; 40/400; 80/400; 160/400).

Naturally infected seeds

Bean seed lots known to be naturally infected with *M. phaseolina* were obtained in July 2015 from different fields in the major bean producing regions in Brazil. Each sample was about 300 g. The samples were enclosed in paper bags with proper labeling and kept in the refrigerator at 5 ± 1 °C until used for subsequent studies. The bean seed lots were previously detected by blotter test (ISTA 1976) for fungal pathogens (samples 400 seeds).

DNA extraction from seed samples

Seed samples (400 seeds per treatment) were ground into a fine powder using a Smart Grind Coffee Grinder (Black & Decker Inc.) prior to DNA extraction. To prevent cross-contamination of DNA, the stainless steel grinding chamber of the grinder was sanitized between isolations by dipping in 1% NaHClO and 70% ethanol for 5 s and subsequently was rinsed in sterile distilled water and dried using sterilized filter paper. Two subsamples (1 g each) per contamination level

Table 1 List of *Macrophomina phaseolina* and other fungal pathogens used in this study

Species	Isolate	Host	Geographic origin	LAMP Result ^b
<i>Macrophomina phaseolina</i>	Mph 22	<i>Phaseolus vulgaris</i>	Unai, MG, Brazil	+
<i>Macrophomina phaseolina</i>	15844	<i>Phaseolus vulgaris</i>	Palotina, PR, Brazil	+
<i>Macrophomina phaseolina</i>	11071	<i>Phaseolus vulgaris</i>	Luiz Eduardo Magalhães, BA, Brazil	+
<i>Macrophomina phaseolina</i>	11219	<i>Phaseolus vulgaris</i>	Primavera do Leste, MT, Brazil	+
<i>Macrophomina phaseolina</i>	3721	<i>Phaseolus vulgaris</i>	Jussara, GO, Brazil	+
<i>Macrophomina phaseolina</i>	14804	<i>Zea mays</i>	Jataí, GO, Brazil	+
<i>Macrophomina phaseolina</i>	11148	<i>Zea mays</i>	Rio Verde, GO, Brazil	+
<i>Macrophomina phaseolina</i>	16731	<i>Glycine max</i>	Costa Rica, MS, Brazil	+
<i>Macrophomina phaseolina</i>	15805	<i>Glycine max</i>	Riachão das Neves, BA, Brazil	+
<i>Macrophomina phaseolina</i>	14138	<i>Glycine max</i>	Sorriso, MT, Brazil	+
<i>Macrophomina phaseolina</i>	13869	<i>Glycine max</i>	Cristalina, GO, Brazil	+
<i>Fusarium oxysporum</i> f. sp. <i>phaseoli</i>	FoxMG14	<i>Phaseolus vulgaris</i>	UFG ^a	—
<i>Sclerotinia sclerotiorum</i>	SPS	<i>Phaseolus vulgaris</i>	UFG ^a	—
<i>Rhizoctonia solani</i>	RS1431	<i>Phaseolus vulgaris</i>	UFG ^a	—
<i>Trichoderma harzianum</i>	TH180	—	UFG ^a	—

^a UFG - Mycological Collection of the Federal University of Goiás, Brazil

^b LAMP result: +: amplification; -: no amplification. A sample was considered positive only if it exhibited a characteristic sky blue color and typical ladder-like pattern on gel electrophoresis. The experiment was repeated twice with similar results

were used for DNA extraction using the CTAB method (Doyle and Doyle 1987). Quality and concentration of DNA were tested using a NanoDrop ND-1000 Spectrophotometer (Thermo Fischer Scientific, Wilmington, DE, USA), followed by 1% agarose gel electrophoresis. The final concentration was adjusted to 10 ng.μL⁻¹ with TE buffer (10 mM Tris-HCl, pH 8, 0.1 mM EDTA). Nucleic acids were stored at -20 °C.

Seed-soaking

Bean seeds were soaked in sterile saline solution (0.85% NaCl, 0.02% Tween 20) at a ratio of 1 seeds.0.1 mL⁻¹. The packages were stored at 4 °C, and 1 mL of seed wash was collected at 2, 4 and 6 h incubation. The total seed wash was centrifuged at 5000 g for 10 min immediately after incubation had finished. The pellet was resuspended in 100 μL sterile milli-Q-water. These aliquots were stored at -20 °C until use. The negative control seed lots consisted of 400 *M. phaseolina*-free seeds. Three replicate subsamples (1 mL) were tested for each sample and the experiment was repeated twice.

Primer design

Macrophomina phaseolina SCAR marker genomic sequences (GU018142.1) generated by Babu et al. (2011) were used as template for primer design applying the Explorer software package (version 4.0) available on the Eiken Chemical Co. Ltd. home page at <http://primerexplorer.jp/elamp4.0.0/index.html>.

The structure of the LAMP primers and their complementarity to target DNA used in this study are shown in Fig. 1. A forward inner primer (FIP) consisted of F1c and F2, and a backward inner primer (BIP) consisted of B1c and B2. The outer primers F3 and B3 were required for initiation of the LAMP reaction. Information regarding the primer names and sequences are provided in Table 2. Each primer was tested in silico for similarities with other sequences available in the GenBank database (<http://www.ncbi.nlm.nih.gov/genbank/>) using the BLASTN algorithm (Altschul et al. 1997).

To confirm the specificity of LAMP amplification for the target sequences, the F3/B3 amplification products (PCR) were sequenced using the Big Dye Terminator v 3.1 Cycle Sequencing Kit (Applied Biosystems, Carlsbad, CA, USA). The automated capillary electrophoresis sequencing runs were performed on an ABI prism 3130 analyzer (Applied Biosystems, Carlsbad, CA, USA). The sequences obtained were compared against the GenBank non-redundant (nr) database using the BLASTx algorithm available at the National Center for Biotechnology Information (<http://www.ncbi.nlm.nih.gov>).

LAMP assay

The LAMP reaction was performed in a total volume of 25 μL: 1.6 μM each of the forward inner primer (FIP) and the backward inner primer (BIP), 0.2 μM each of the forward outer primer (F3) and the backward outer primer (B3), 16 U of Bst DNA polymerase (New England Biolabs), 1.4 mM each

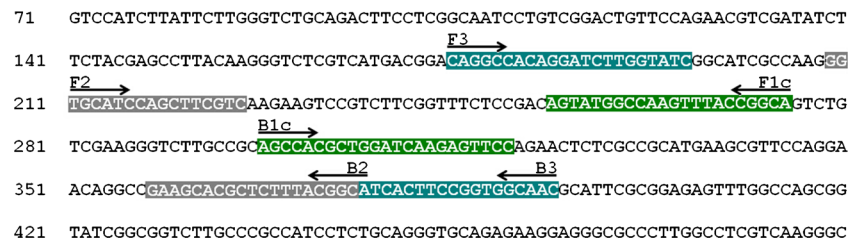


Fig. 1 Design of LAMP primers for detection of *M. phaseolina*. The partial SCAR marker genomic sequence (GU018142.1) used for LAMP primers and the location of the four primers, FIP (F1c-F2), BIP (B1c-B2c), F3 and B3c are shown. The arrows indicate the direction of extension

of the dNTPs (Thermo Fischer Scientific, Wilmington, DE, USA), 10 mM of KCl and $(\text{NH}_4)_2\text{SO}_4$, 20 mM of Tris-HCl buffer (pH 8.8), 0.8 M of Betaine (Sigma-Aldrich, St. Louis, MO-Aldrich), and 8 mM of MgSO_4 were mixed. To identify the optimal temperature and time for the visual detection of LAMP amplification, the reactions were performed at 60 °C, 63 °C and 65 °C for 45 and 60 min then heated for 10 min at 80 °C to terminate the reaction using a Veriti Thermal Cycler (Applied Biosystems, Carlsbad, CA, USA), as a heating block. Additional detection reagent was adjusted to reach a final concentration of 120 μM hydroxynaphthol blue (HNB) (Sigma-Aldrich, St. Louis, MO Aldrich).

Template DNA was replaced by Milli-Q-water as negative control. The amplification product was visually inspected according to the color change from violet to sky blue, while the negative control remained violet. The tubes were observed by naked eyes and photographed under the natural light. In addition, each product was analyzed by 1.0% agarose gel electrophoresis stained with ethidium bromide and photographed under a UV transilluminator. Two replicate were tested for each sample.

Primer specificity and sensitivity

Template DNA was prepared as described earlier and was 10-fold serially diluted from 10 $\text{ng} \cdot \mu\text{L}^{-1}$ to 10^{-4} $\text{ng} \cdot \mu\text{L}^{-1}$ prior to use as the template in LAMP assay. The process and conditions were similar to the LAMP assay described as above. When the LAMP reactions were completed, the products were analyzed on HNB-visualized color change and 1% agarose gel electrophoresis. The test was performed twice.

Primer specificity was tested individually on *M. phaseolina* isolates collected from different areas of Brazil in 2015 and

other important plant-pathogenic fungi in agricultural production, including *Sclerotinia sclerotiorum*, *Fusarium oxysporum* f. sp. *phaseoli*, *Rhizoctonia solani* and *Trichoderma harzianum* obtained from the Federal University of Goiás Collection (Table 1).

Statistical analysis

Results (mean for positive subsamples replications of each DNA sample) were subject to analysis of variance (ANOVA) with the samples means compared using the Tukey's method at a probability of 5% using R software.

Results

Lamp primer sensitivity and specificity test

The reaction mixture was incubated at 60 °C, 63 °C, or 65 °C for 45 and 60 min to establish the best LAMP assay conditions for the primer set. Positive results were obtained for all conditions, and no differences were observed in the amplification efficiency based on HNB color change or the ladder-like pattern of the LAMP products on agarose gel. Thus, based on stringency conditions and time, we established the ideal conditions for the LAMP assay to use incubation at 65 °C for 45 min.

The sensitivity of this LAMP assay for *M. phaseolina* was evaluated using the optimal reaction condition and ten-fold, serially diluted DNA from *M. phaseolina*. The detection limit was 1 μg of genomic DNA per reaction (25 μL) whether the detection involved HNB or gel electrophoresis (Fig. 2a).

Table 2 List of primers used during the current study

Primer type	Sequence (5'-3')	Length
MphLF3	CAGGCCACAGGATCTTGGTATC	22
MphLB3	GTTGCCACCGGAAGTGAT	18
MphLFIP	TGCCGGTAAACTTGGCCATACTGGTGCATCCAGCTTCGTC	40
MphLBIP	AGCCACGCTGGATCAAGAGTTCGCCGTAAGAGCGTGCTTC	41

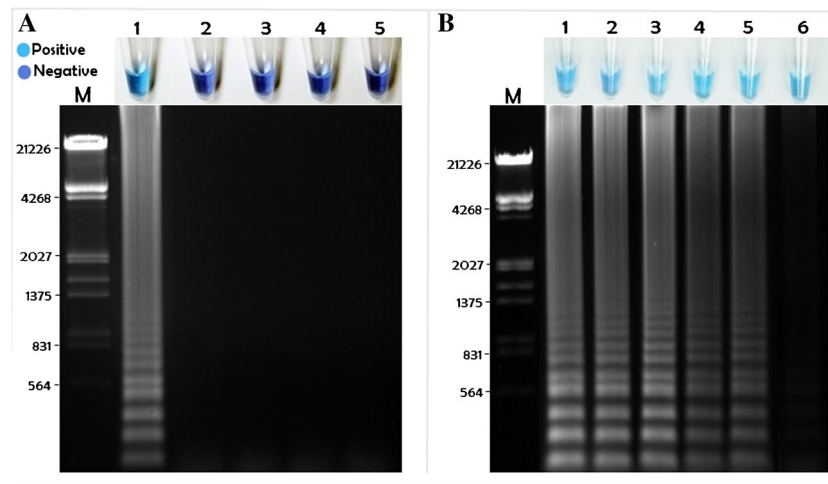


Fig. 2 Sensitivity and Specificity of LAMP for detection of *M. phaseolina* genomic DNA. Detection by LAMP and HNB visualization of colour change or gel electrophoresis analysis of the LAMP products. **M** indicates a Lambda DNA/EcoRI plus Hind III marker (Thermo Fischer Scientific); **a** Sensitivity: Concentration of

template DNA ($\text{ng} \cdot \mu\text{L}^{-1}$) per reaction was: **1**–10, **2**–1, **3**– 1.10^{-1} , **4**– 1.10^{-2} , **5**– 1.10^{-3} , **6**– 1.10^{-4} . **b** Specificity: DNA Sample **1** - *M. phaseolina*; **2** - *S. sclerotiorum*; **3** - *F. oxysporum* f. sp. *Phaseoli*; **4** - *R. solani*; **5** - *T. harzianum* (see Table 1)

The specificity of the LAMP assay was tested using DNA from five other fungal species (*Sclerotinia sclerotiorum*, *Fusarium oxysporum* f. sp. *phaseoli*, *Rhizoctonia solani*, and *Trichoderma harzianum*) which are also pathogens frequently associated with common bean seeds (Table 1). Samples containing *M. phaseolina* DNA developed a characteristic sky-blue color after the reaction. Samples with no fungal DNA, or containing DNA from other fungal pathogens, instead remained violet under ambient light (Fig. 2b). The results of the HNB staining method were reconfirmed when the LAMP products were subjected to gel electrophoresis (Fig. 2b).

To maximize the genetic diversity among the *M. phaseolina* isolates tested, ten isolates collected from the major common bean-producing regions in Brazil were included in the specificity test. The results showed that the primer sets were effective, and the same results were obtained with HNB visualization and gel electrophoresis for all isolates (Table 1).

The specificity of the LAMP region in all isolates was confirmed by sequencing using primers F3 and B3. The amplification products from PCR corresponded with the selected target sequence.

Evaluation of the LAMP assay using common bean seed DNA

The usefulness of the LAMP methods developed during the current study was evaluated by performing an analysis of seed lots. Uninfected common bean seeds were mixed with *M. phaseolina*-inoculated seeds in order to model different infection levels of (0%–40%).

Detection with the LAMP assay was possible in samples containing 0.25% infection incidence, corresponding to one

infected seed among 399 uninfected seeds (Fig. 3a). The results of the HNB staining method were reconfirmed by subjecting LAMP products to gel electrophoresis (Fig. 3a). Negative results were obtained with uninfected seeds (0/400 seeds).

Three bean seed lots harvested from *M. phaseolina*-infected fields were evaluated (21.038, 21.039, and 21.040). According to the blotter test analyses, the mean fungal incidence rates in these lots averaged 10.0%, 11.5%, and 10.5%, respectively, for these lots, using samples of 400 seeds for each lot. The LAMP assay was able to detect *M. phaseolina* in samples from all of these seed lots (Fig. 3b). The results of the HNB staining method were reconfirmed using gel electrophoresis of the LAMP products (Fig. 3b). In all, two replicate subsamples using DNA extraction were tested for each seed lot, along with two LAMP reactions, for a total of 4 subsamples per lot sample. All four subsamples were positive (data not shown), indicating a sensitivity of 100% for the LAMP assay.

Evaluation of the LAMP assay using seed-soaking

The LAMP assay was also assayed for its ability to detect *M. phaseolina* in soaked seeds (direct-LAMP), which does not require prior fungal DNA extraction.

In order to set the conditions for direct-LAMP, an artificial infection level of 1% was created by mixing four *M. phaseolina*-inoculated seeds with 396 uninfected seeds. Then, the seeds were soaked in a sterile saline solution, at a ratio of 1 seed: 0.1 mL^{-1} and three aliquot subsamples were taken during the first 6 h, at 2, 4, and 6 h. Three replicate subsamples were tested for each seed lot sample with, along with two replicates for the LAMP reaction, for a total of 6 subsamples per sample time point. The percentages of positive

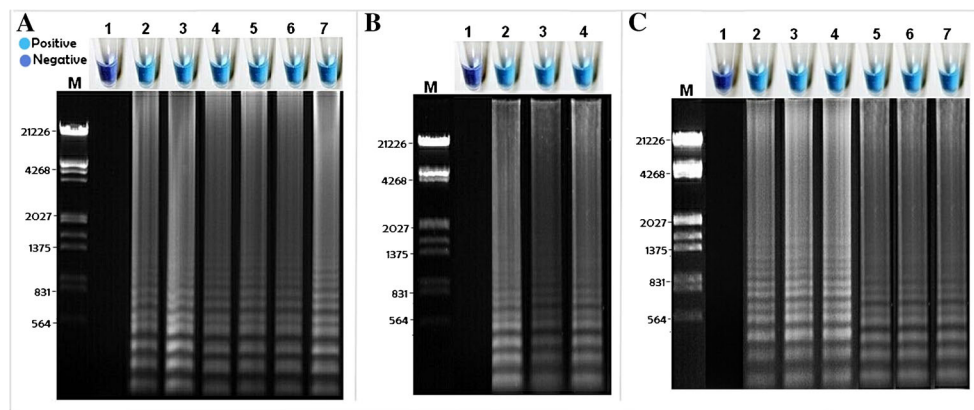


Fig. 3 LAMP in detection of seeds lots with different infection levels. Analysis based on detection by LAMP and HNB visualization of color change or gel electrophoresis analysis of the LAMP products. **M** - Indicates a Lambda DNA/EcoRI plus Hind III marker (Thermo Fischer Scientific); **(a)** Sensitivity: Uninfected common bean seeds were mixed with *M. phaseolina* inoculated seeds in order to model different infection levels in 400 seeds. Sample 1–0%; 2–0.25%; 3–0.5%; 4–1%; 5–10%; 6–20%; 7–40%. **(b)** Validation of LAMP detection in naturally infected

seeds: Seed lots harvested from *M. phaseolina*-infected fields. Sample 1- negative control (0%), 2- lot 21.038; 3- lot 21.039; 4- lot 21.040; **(c)** Validation of LAMP detection in seed-soaking: Sample 1- negative control (0%), 2–4: Seed lot 21.038, 21.039 and 21.040 - contained incidence averaged 10.0%, 11.5%, and 10.5% *M. phaseolina* infected seeds, respectively; 2 h soaking in sterile saline solution. 5–7: Artificially inoculated seed – 1%; 2, 4 and 6 h soaking in sterile saline solution, respectively

subsamples were compared to measure the effects of seed-soaking duration on detection sensitivity.

With an artificial infection level of 1%, it was observed that the direct-LAMP detected *M. phaseolina* in at least one subsample of the three collected at each sample time point, giving a mean sensitivity of 33% for detecting positive subsamples. Interestingly, no significant differences ($P < 0.05$) were observed among the mean sensitivity percentages calculated using positive subsamples at the 2, 4, or 6 h soaking times.

These results were confirmed with naturally infected seeds (lots 21.038, 21.039, and 21.040), equivalent to 10%–11% *M. phaseolina*-infected seeds according to the previous blotter test analyses. *M. phaseolina* was detected in all of the samples following 2 h soaking (Fig. 3c). The direct-LAMP method identified a higher proportion (5/6) of positive subsamples and thus showed improved detection, for a sensitivity of 83%. Analysis of variance followed by Tukey's honest significant difference test for pairwise comparisons showed that for these primer sets, there were no significant differences ($P < 0.05$) between the percentages of positive subsamples detected in each seed lot sample. Negative LAMP reactions were observed in uninfected common bean seeds.

However, significant differences for the sensitivity of percentages of positive subsamples detected by direct-LAMP were found for different inoculum levels. Although has been possible the detection of the *M. phaseolina* in all seed lots tested, the results showed that the frequency of detection, as determined by percentages of positive subsamples, in seed lots with $\leq 1\%$ infection incidence was lower than in the lots with $\geq 10\%$ infection incidence.

Discussion

In order to facilitate management of *M. phaseolina* via early, sensitive detection methods that could be used with seeds, we developed a LAMP assay with primers specific for the DNA conserved regions (SCAR-GU018142.1) of the *M. phaseolina*. The use of genomic marker sequences of unknown function and localization as sequence sources for primer design has been used to develop LAMP assays to monitor and detect fungal pathogens in the field, such as *P. sojae* (Dai et al. 2012), *S. sclerotiorum* (Duan et al. 2014a, b) and *Verticillium dahlia* (Moradi et al. 2014). However, with *M. phaseolina*, no such approach had been described to date.

The *M. phaseolina*-specific LAMP assay reported here demonstrated high test specificity and sensitivity. Compared with conventional methods, the LAMP method has multiple advantages. First, LAMP can be used with a reaction time of 45–60 min to completion, compared to 2–10 days for incubation methods. Also, culturing is not practical for the rapid detection of *M. phaseolina*, and due to wide variations in the phenotype of the isolates, this criteria is often unreliable (Babu et al. 2011). Secondly LAMP has a higher sensitivity than conventional methods of detection, allowing for the identification of one infected seed in a lot of 400 seeds.

The detection limit obtained with our LAMP primer set designed to detect *M. phaseolina* genomic DNA (1 pg) was found to be higher than the limit obtained by ITS-Mp-LAMP assays (100 pg) designed for internal transcribed spacers sequence and used to detect *M. phaseolina* in diseased host tissues (Lu et al. 2015).

Several published reports demonstrate that LAMP amplifications tolerate higher levels of certain inhibitors and that the sensitivity is less affected by the various components of the

plant samples, allowing for the omission of sample DNA purification (Kaneko et al. 2007; Francois et al. 2011). In the present study, a positive LAMP reaction was obtained in all samples after 2–6 h of seed-soaking without significant inhibition, showing that direct-LAMP was tolerant to seed sample-derived inhibitors. This is especially advantageous when large numbers of samples need to be tested. In addition, the tolerance of LAMP for seed components obviates the need for pre-extraction of the DNA template, reducing the time needed for the entire process. The conventional protocol (CTAB-based DNA extraction) takes approximately one hour to prepare high quality DNA. This can be exploited to develop rapid extraction techniques suitable for simple field-based qualitative tests.

Although the direct-LAMP assays in this study showed high specificity, we still need to be cautious because of the low detection sensitivity, i.e., the percentages of positive subsamples, observed in seed lots with low levels of infection ($\leq 1\%$). Clearly, the number of seeds required, as well as the number of subsamples needed, must be optimized to improve the sensitivity. Thus, further studies need to be carried out on a larger scale, using more samples of bean seed lots harvested from *M. phaseolina*-infected fields, to confirm the sensitivity of the direct-LAMP assay.

An important advantage of LAMP is that it allows visual detection of amplification by inspection with the naked eye, and requires only simple equipment, such as a heating block. In this study, the positive and negative reactions could be successfully distinguished with the naked eye by adding hydroxynaphthol blue, a colorimetric indicator of calcium and alkaline earth metal ions. In a LAMP reaction mixture, dNTPs can influence the color of hydroxynaphthol blue by chelating with Mg^{2+} ions. In the presence of hydroxynaphthol blue, the color gradually changes from violet to sky blue due to the decrease in the magnesium concentration as it binds to pyrophosphate during amplification (Goto et al. 2009).

The LAMP products can be also quantitatively assessed by measuring the fluorescence intensity from DNA-intercalating fluorophores such as SYBR Green I. However, the use of DNA-binding dyes after the reaction endpoint increases the risk of cross-contamination from open cover operations (Notomi et al. 2000; Zhang et al. 2014). To avoid such contamination, a visualization indicator (HNB) is applied in the LAMP assay prior to amplification.

In addition, the turbidity caused by the accumulation of magnesium pyrophosphate precipitate can be measured by a real-time turbidimeter (Mori et al. 2001, 2004; Mori and Notomi 2008; Goto et al. 2009; Tomlinson et al. 2010; Zhang et al. 2013; Zhang et al. 2014). However, the use of expensive equipment greatly limits the wide use of this procedure in agricultural extension stations.

In this context, and considering the high sensitivity obtained in detecting *M. phaseolina* DNA in seed samples, our

LAMP assay could be potentially useful for screening seed lots with either extracted DNA or seed-soaking, which would greatly reduce the risks of planting seed lots with low frequencies of infection.

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